

UTQIAGVIK NSA, AK, USA

WMO: 700270

Lat: 71.323N

Lon: 156.616W

Elev: 8

StdP: 101.23

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 27516

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-38.6	-36.1	-40.1	0.1	-36.8	-38.2	0.1	-35.0	12.5	-19.4	11.4	-18.0	0.3	N/A	1.039

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.1	13.4	10.9	11.4	9.8	9.6	8.4	11.2	13.0	9.9	11.4	8.5	9.7	3.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	7.7	12.3	8.8	7.0	10.9	7.5	6.4	9.5	32.0	13.2	28.9	11.3	25.9	9.7	14.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.0	8.9	DB	-40.6	18.3	2.9	2.5	-42.8	20.0	-44.5	21.5	-46.1	22.8	-48.3	24.6	(4)
(5)			WB	-39.2	13.7	2.6	1.1	-41.1	14.5	-42.7	15.1	-44.1	15.8	-46.0	16.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.9	-23.9	-23.7	-24.1	-15.9	-5.3	1.3	4.9	4.2	1.4	-4.4	-14.1	-20.7		
	DBStd	12.25	6.95	7.41	6.43	6.09	4.51	2.87	3.18	3.01	2.75	3.58	6.11	6.00		
	HDD10.0	7287	1052	944	1056	776	476	262	162	182	259	445	723	950		
	HDD18.3	10322	1311	1178	1314	1026	734	512	416	438	509	703	973	1208		
	CDD10.0	7	0	0	0	0	0	0	4	3	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	3.7	2.0	2.4	3.0	3.8	4.4	4.4	4.5	4.3	4.4	5.2	4.0	2.1		
(15) Precipitation	PrecAvg	112	3	3	3	3	5	8	22	26	18	12	6	4		
	PrecMax	156	16	11	6	18	11	24	54	67	48	38	12	9		
	PrecMin	45	0	0	0	0	0	0	1	5	4	4	2	1		
	PrecStd	31	4	3	2	4	3	6	14	16	11	8	3	2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.5	-2.5	-3.5	1.0	2.4	13.5	17.5	15.5	11.7	3.3	0.4	-2.2		
		MCWB	-1.9	-2.9	-4.1	0.0	1.4	10.1	13.1	11.9	9.7	2.5	-0.3	-2.6		
	2%	DB	-5.5	-5.7	-7.3	-2.5	1.4	10.0	13.8	13.1	8.8	1.7	-1.3	-5.8		
		MCWB	-5.9	-6.2	-7.7	-2.9	0.7	8.0	11.4	10.9	7.7	0.8	-1.8	-6.3		
	5%	DB	-11.4	-9.4	-11.7	-4.7	0.7	7.2	11.6	11.2	6.9	0.7	-3.2	-8.6		
		MCWB	-11.8	-9.8	-12.0	-5.0	0.2	5.7	10.1	9.8	6.3	-0.1	-3.9	-8.9		
	10%	DB	-14.9	-13.0	-15.0	-7.2	0.2	5.1	9.9	9.1	5.1	0.0	-5.2	-11.5		
		MCWB	-15.2	-13.3	-15.4	-7.6	-0.2	4.0	8.8	8.1	4.5	-0.6	-5.8	-11.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-2.2	-2.6	-0.7	1.9	10.8	13.9	12.2	9.7	3.2	0.2	-2.4		
		MCDB	-1.0	-1.9	-2.0	0.2	2.2	13.8	16.9	14.1	11.5	4.0	0.6	-2.1		
	2%	WB	-5.3	-4.8	-6.6	-3.6	1.1	8.2	11.8	10.8	7.9	1.5	-1.1	-6.9		
		MCDB	-4.8	-4.4	-6.1	-3.3	1.4	10.2	13.5	12.4	8.7	1.8	-0.6	-6.5		
	5%	WB	-10.3	-7.3	-9.8	-5.7	0.6	6.2	10.4	9.2	6.3	0.5	-3.0	-9.2		
		MCDB	-9.8	-6.8	-9.4	-5.4	0.9	7.8	12.0	10.2	6.9	1.0	-2.4	-8.9		
	10%	WB	-14.1	-10.9	-14.0	-8.2	0.1	4.4	9.3	7.6	4.7	-0.2	-4.6	-12.6		
		MCDB	-13.8	-10.4	-13.7	-7.9	0.4	5.5	10.6	8.3	5.1	0.4	-4.0	-12.3		
(35) Mean Daily Temperature Range	MDBR		6.5	6.8	6.8	6.8	3.9	4.1	5.1	3.9	3.0	3.7	5.7	6.4		
	5% DB	MCDBR	11.4	11.1	9.3	6.9	3.7	9.0	8.9	8.2	6.3	3.6	6.6	8.4		
		MCWBR	11.1	11.0	9.2	5.8	3.1	7.1	6.7	5.9	4.6	3.3	6.7	8.5		
	5% WB	MCDBR	11.9	11.2	8.8	5.9	3.2	9.5	8.6	7.7	5.4	3.8	6.3	8.5		
		MCWBR	11.8	11.0	8.8	5.9	3.0	7.5	6.9	6.1	4.4	3.6	6.3	8.3		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.201	0.244	0.305	0.315	0.312	0.333	0.329	0.284	0.202	N/A	N/A		
	taud		N/A	1.971	2.155	2.049	2.127	2.296	2.418	2.520	2.555	2.135	N/A	N/A		
	Ebn at Noon		0	467	753	802	851	872	838	798	739	476	0	0		
	Edh at Noon		0	35	69	112	119	105	90	72	50	27	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.00	0.27	1.84	4.09	5.57	6.47	5.07	2.79	1.32	0.32	0.00	0.00		
	RadStd		0.00	0.03	0.11	0.27	0.46	0.32	0.56	0.28	0.11	0.04	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+1.66	N/A	N/A	N/A	N/A	N/A	-637	-633	N/A	N/A		
(47) Regional (3 neighbors)		+1.46	+2.68	+1.38	N/A	N/A	N/A	-483	-485	N/A	N/A		

Nomenclature: See separate page